

XLIH. *On Discontinuous Movements of Fluids.**By* PROFESSOR HELMHOLTZ*.

IT is well known that the hydrodynamic equations give exactly the same partial differential equation for the interior of an incompressible fluid free from friction, and the particles of which do not rotate, as obtains for stationary currents of electricity or heat in conductors of uniform conductivity. It might be expected, therefore, that, for the same shape of the space traversed and for like conditions of boundary, the form of the stream for fluids would be the same as that for electricity and heat, with the exception of trifling differences caused by secondary conditions. In fact, however, the differences in many cases are very evident and profound between the arrangement of the current of a fluid and that of the above-mentioned imponderables.

Such differences are especially obvious if the current enters through an opening with sharp edges, into a wider space. In such cases the radiant lines of electricity separate from one another in all directions immediately on passing the opening; while a fluid current, whether water or air, moves forwards at first from the orifice in a compact stream, which is then usually dissipated in gyrations at a greater or less distance. The parts of the fluid of the larger vessel which lie in the neighbourhood of the orifice, but at the sides of the current, may, on the contrary, remain completely at rest. Every one is acquainted with this sort of motion, which is shown clearly by a current of air impregnated with smoke. In fact in these phenomena the compressibility of the air is of little influence; and air shows, with very few deviations, the same forms of motion as water.

Such great differences between what actually takes place and the deductions from the theoretical analysis hitherto accepted, must cause physicists to regard the hydrodynamical equations as a practically very imperfect approximation to the truth. The cause of this discrepancy might be supposed to lie in the internal friction of the fluid, although the divers strange and saltatory irregularities which every one has encountered who has experimented upon the motions of fluids can in no wise be accounted for by the continuous and uniform action of friction.

The investigation of the case where periodical motions result from a continuous air-current, as, for instance, in organ-pipes, convinced me that such an action could only arise by a discontinuous motion of the air, or at least by an approximately discontinuous one. Hence I was led to the discovery of a condi-

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tion which must be taken into account in the integration of the hydrodynamic equations, and which has, as far as I am aware, been hitherto neglected. By taking this condition into account, on the other hand, where the calculation can be carried out, forms of motion are obtained such as are observed in reality. The circumstance in point is the following:—

In the hydrodynamic equations the velocity and the pressure of the current-particles are treated as continuous functions of the coordinates. But, on the other hand, if we consider a perfect liquid (that is, one not subject to friction), there is nothing essential in its nature to prevent two immediately neighbouring liquid layers from slipping past one another with finite velocity. At all events, those properties of fluids which are considered in the hydrodynamic equations, namely the constancy of the mass in all units of volume and the equality of the pressure in all directions, clearly offer no obstacle to the occurrence of tangential motions of finite difference of magnitude on both sides of a plane passing through the interior. The components perpendicular to the surface of the velocity and the pressure must, on the other hand, of course be equal on both sides of such a surface. In my work on gyratory movements*, I have already pointed out that such a case must occur if two previously separated masses of water come into superficial contact when they have different motions. In my above-mentioned work I was led to the idea of such a *surface of separation*, or, as it was then called, "*gyration-surface*," by the conception of lines of gyration arranged continuously along a surface, the mass of which may be vanishingly small, while its moment of rotation remains finite.

Now, in a fluid which at first is at rest, or in continuous motion, a finite difference of motion between immediately neighbouring particles can only be brought about by a moving force which acts discontinuously. Of external forces, we here only consider the force of impact.

But there exists also in the interior of the fluids a cause which may give rise to discontinuity in the motion. Pressure, indeed, may assume any positive value, and the density of the fluid will vary continuously with the pressure. But the moment the pressure passes zero and commences to become negative, a discontinuous change in the density takes place, the fluid is broken asunder.

Now the magnitude of the pressure in a moving fluid depends upon the velocity, and, indeed, in incompressible fluids the diminution of the pressure under otherwise similar circumstances is directly proportional to the *vis viva* of the moving particles. If the *vis viva* exceeds a certain amount, the pressure must in fact

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become negative, and the fluid must be torn apart. At such a place the accelerating force which is proportional to the differential coefficient of the pressure becomes, of course, discontinuous, and consequently the condition is fulfilled which is necessary to bring about a discontinuous motion in the fluid. The motion of the fluid in passing by such a place must be of such a nature that a surface of separation is there formed.

The velocity necessary to bring about the rupture of the fluid is that which the fluid would assume if it were to flow into *vacuum* under the pressure which the fluid would have at the same place when at rest. This is certainly a comparatively considerable velocity; but we must bear in mind that if fluids are to flow continuously like electricity, the velocity at every sharp edge around which the current turns must be infinitely great*. It consequently follows that *every perfect geometrically sharp edge by which a fluid flows must tear it asunder and establish a surface of separation, however slowly the rest of the liquid may move.* When the edges are imperfectly formed, and rounded, such a separation requires a certain increased velocity. Pointed projections on the side of the orifice must act in the same manner.

In regard to gases, the same condition obtains with them as with liquids, with the exception that the *vis viva* of the motion of a particle is not directly proportional to the diminution of the pressure p , but to the quantity p^m , where, taking into account the cooling of the air by expansion, $m = 1 - \frac{1}{\gamma}$, γ being the ratio between the specific heat with constant pressure and that with constant volume. For atmospheric air, the exponent m has the value 0.291. Since this is positive and real, p^m and p can only decrease to nothing, and not become negative as the velocity becomes high. It would be otherwise if gases simply followed Mariotte's law, and did not suffer any change of temperature. The value $\log p$ would then enter instead of p^m ; and this value may become infinite and negative without p becoming negative. Under this condition, a tearing asunder of the mass of air would not be necessary.

We may satisfy ourselves of the actual existence of such discontinuities by allowing a stream of air impregnated with smoke to flow through a round hole or cylindrical tube with moderate velocity so that no hissing ensues. Under favourable circumstances thin streams may in this way be obtained of a line in diameter and several feet long. Within the cylindrical

* At the very small distance ϱ from a sharp edge the faces of which are inclined at an angle α , the velocity becomes infinite with ϱ^{-m} , where

$$m = \frac{\pi - \alpha}{2\pi - \alpha}.$$

surface the air is then in motion with constant velocity; but outside the stream, even quite close to it, the air is not at all, or scarcely at all, moved. This sharp separation may also be seen very distinctly if a calmly flowing cylindrical current of air be led through the apex of a flame. A sharply bordered piece of the flame is cut out, while the rest of it remains quite undisturbed, or at most a very thin sheet, corresponding to the boundary layers caused by the friction, is carried along a little way with the air-current.

With regard to the mathematical theory of these motions, I have already given the limiting conditions for an internal surface of separation of the liquid. These conditions are, that the pressure on both sides of the surface must be the same, as also must be the components of the velocity in a direction perpendicular to the surface of separation. Since, now, the motion throughout the interior of an incompressible fluid, the particles of which have no rotary motion, is fully known when the motion of its entire surface and its interior discontinuities are given, it follows generally that when a fluid is rigidly enclosed, we have only to consider the motion of the surface of separation and the changes in the discontinuity.

Such a surface of separation can be treated mathematically exactly as if it were a surface of gyration—that is, as if it were completely covered with lines of gyration (*Wirbelfäden*) of infinitely small mass, but finite moments of rotation. In every surface unit of such a surface of gyration there is one direction in which the components of the tangential velocities are equal. This gives at once the direction of the lines of gyration at the corresponding place. The moment of these lines must be made proportional to the difference shown by the perpendicular components of the tangential velocity on both sides of the surface.

The existence of such lines of gyration for an ideal fluid without friction is a mathematical fiction which facilitates the integration. In an actual liquid subjected to friction this fiction is quickly realized, because the bordering particles are set in rotation by the friction, and then immediately are formed lines of gyration of finite gradually increasing mass, while the discontinuity of the motion is thereby at the same time equalized.

The motion of a surface of gyration, and of the lines of gyration lying in it, are to be determined according to the rules established in my work on gyratory motion. The mathematical difficulties of this problem can only, it is true, be overcome in a few of the more simple cases. But in many other cases we may at least draw conclusions concerning the direction of the changes by means of the method of viewing the matter just described.

It is especially to be remarked that, according to the laws

established for gyratory motions, the lines, and with them the surfaces of gyration, can neither originate nor disappear in the interior of a fluid free from friction, but that, on the contrary, every line of gyration must retain the same moment of rotation constant. It is further clear that the lines of gyration upon a gyration surface must advance with a velocity which is the mean of the velocities existing on both sides of the surface. It hence follows that a surface of separation can only be elongated in that direction in which the stronger of the two contiguous currents in it moves.

I sought, in the first place, for examples of persistent surfaces of separation in stationary currents which would allow of integration, in order to test whether the theory gives forms of currents which correspond better with experiment than when the discontinuity of the motion is neglected. If a surface of separation which divides flowing from still water has to remain stationary, the pressure along the surface must be the same in the moving layer as in the stationary one; whence it follows that the tangential velocity of the particles of water throughout the whole extent of the surface must be constant. So also must be the density of the fictitious lines of gyration. The beginning and end of such a surface can only lie at the side of the vessel, or at an infinite distance. In the first case they must be tangential to the side of the vessel, provided that the latter be continuously bent, because the component of the velocity perpendicular to the side of the vessel must be equal to nothing.

The stationary forms of the surfaces of division are distinguished, as experiment and theory alike indicate, by a remarkably high degree of alterability when subjected to the least disturbance, so that they comport themselves in some degree like bodies in unstable equilibrium. The remarkable sensitiveness to sound of a cylindrical current of air impregnated with smoke has already been described by Dr. Tyndall. I have verified Dr. Tyndall's experiments. This is clearly a property of the surfaces of separation, which is of the greatest importance in sounding musical pipes.

Theory points out that, wherever an irregularity is formed on the surface of an otherwise stationary current, this must give rise to a progressive spiral unrolling of the corresponding part of the surface; the corresponding portion, moreover, advances along the current. This endeavour towards spiral unrolling at every interruption is also easily recognizable on observing the currents. According to theory, a prismatic or cylindrical current could be infinitely long. But in practice such a current cannot be formed, because in an element which is so easily moved as air it is impossible entirely to avoid small disturbances.

It is easy to see that the conditions of the state of rest are satisfied by such an infinitely long cylindrical current which, issuing out of a tube of the same diameter, enters into a still external fluid, and contains throughout fluid which moves parallel to its axis with uniform velocity.

I shall here merely give a sketch of the mathematical treatment of a case of the inverse kind, where the current passes out of a wide space into a narrow channel, in order at the same time to give an example of a method by which certain problems can be solved in the doctrine of potential functions which have hitherto presented difficulty.

I confine myself to the case where the motion is "stationary," and depends alone upon two rectangular coordinates x, y , and where, further, in the frictionless fluid there are no rotating particles, and consequently none such can arise. For the particle of fluid at the point (x, y) , let us denote by u the component velocity parallel to x , and by v that parallel to y . Then two functions of x and y may, as is well known, be found such that

$$\left. \begin{aligned} u &= \frac{d\phi}{dx} = \frac{d\psi}{dy}, \\ v &= \frac{d\phi}{dy} = -\frac{d\psi}{dx}. \end{aligned} \right\} \dots \dots \dots (1)$$

By means of these equations the condition is at once fulfilled in the interior of the fluid, that the mass remains constant for every unit of volume, namely

$$\frac{du}{dx} + \frac{dv}{dy} = \frac{d^2\phi}{dx^2} + \frac{d^2\phi}{dy^2} = \frac{d^2\psi}{dx^2} + \frac{d^2\psi}{dy^2} = 0. \dots \dots (1a)$$

With the constant density h , and when the power of the external force is represented by V , the pressure in the interior of the fluid is given by the equation

$$\left. \begin{aligned} V - \frac{p}{h} + C &= \frac{1}{2} \left[\left(\frac{d\phi}{dx} \right)^2 + \left(\frac{d\phi}{dy} \right)^2 \right] \\ &= \frac{1}{2} \left[\left(\frac{d\psi}{dx} \right)^2 + \left(\frac{d\psi}{dy} \right)^2 \right] \end{aligned} \right\} \dots \dots (1b)$$

The curves

$$\psi = \text{const.}$$

are the current lines of the fluid, and the curves

$$\phi = \text{const.}$$

are orthogonal to them. The latter are the curves of equal power in the case of electricity, or of equal temperature in the case of heat, flowing in a stationary current in conductors of constant conductivity.

From equation (1) we get as integral equation that the value $\phi + \psi i$ is a function of $x + yi$ (where $i = \sqrt{-1}$). The solutions hitherto found generally express ϕ and ψ as a sum of terms which are themselves functions of x and y . But, inversely, we may regard $x + yi$ as a function of $\phi + \psi i$ and develop. In problems dealing with currents between two rigid walls, ψ is constant along the borders; so that if ϕ and ψ be laid down as rectangular coordinates in a plane, we have, in a strip of this plane bounded by two parallel straight lines $\psi = c_0$ and $\psi = c_1$, to seek the function $x + yi$ so that it may correspond at the edge to the equation of the wall, and in the interior assume the given discontinuities.

A case of this kind is when we put

$$x + yi = A \{ \phi + \psi i + e^{\phi + \psi i} \}, \quad . \quad . \quad . \quad (2)$$

or

$$x = A\phi + Ae^{\phi} \cos \psi,$$

$$y = A\psi + Ae \sin \psi :$$

for the value $\psi = \pm \pi$, y is constant, and

$$x = A\phi - Ae^{\phi}.$$

If ϕ varies from $-\infty$ to $+\infty$, x changes at the same time from $-\infty$ to $-A$, and then back again to $-\infty$. The current-curves $\psi = \pm \pi$ correspond, therefore, to the current along two straight walls, for which $y = \pm A\pi$, and x varies between $-\infty$ and $-A$.

If, therefore, we use ψ to express the current-curves, equation (2) corresponds to a current which flows into infinite space from a canal bordered by two parallel planes. At the edge of the canal, however, where $x = -A$ and $y = \pm A\pi$, and, further, where $\phi = 0$ and $\psi = \pm \pi$,

$$\left(\frac{dx}{d\phi} \right)^2 + \left(\frac{dy}{d\phi} \right)^2 = 0,$$

$$\left(\frac{d\phi}{dx} \right)^2 + \left(\frac{d\phi}{dy} \right)^2 = \infty.$$

Electricity and heat may flow in this manner; liquids would be torn asunder.

If from the edge of the canal there should proceed stationary lines of division, which of course would be continuations of the current-lines $\psi = \pm \pi$ passing along the wall, and if exteriorly to these lines of division which border the current fluid there should be rest, then the pressure must be the same on both sides of the lines of division. That is, along those portions of the

lines $\psi = \pm \pi$ which correspond to the independent divisional lines we must have, according to (1b),

$$\left(\frac{d\phi}{dx}\right)^2 + \left(\frac{d\phi}{dy}\right)^2 = \text{const.} \quad . \quad . \quad . \quad (3)$$

In order now to maintain the chief features of the motion given in equation (2), let us add to the above expression for $x + yi$ another term $\sigma + \tau i$, which is also a function of $\phi + \psi i$.

We have then

$$\left. \begin{aligned} x &= A\phi + Ae^{\phi} \cos \psi + \sigma, \\ y &= A\psi + Ae^{\phi} \sin \psi + \tau; \end{aligned} \right\} \quad . \quad . \quad . \quad (3a)$$

and we must determine $\sigma + \tau i$ so that, along the free portion of the divisional surface $\psi = \pm \pi$, we may have

$$\left(A - Ae^{\phi} + \frac{d\tau}{d\phi}\right)^2 + \left(\frac{d\tau}{d\phi}\right)^2 = \text{const.}$$

This condition is fulfilled if, in the same place, we make

$$\frac{d\sigma}{d\phi} = 0, \text{ or } \sigma = \text{const.}, \quad . \quad . \quad . \quad (3b)$$

and

$$\frac{d\tau}{d\phi} = \pm A \sqrt{2e^{\phi} - e^{2\phi}}. \quad . \quad . \quad . \quad (3c)$$

Since ψ is constant along the wall, we can integrate the last equation in regard to ϕ , and convert the integral into a function of $\phi + \psi i$ by putting $\phi + i(\psi + \pi)$ everywhere in place of ϕ . We thus, after properly determining the constants of integration, get

$$\sigma + \tau i = Ai \left\{ \sqrt{-2e^{\phi + \psi i} - e^{2\phi + 2\psi i}} + 2 \arcsin \left[\frac{i}{\sqrt{2}} e^{\frac{1}{2}(\phi + \psi i)} \right] \right\}. \quad (3d)$$

The points of divergence of this expression are where $e^{\phi + \psi i} = -2$, that is, where $\psi = \pm (2n + 1)\pi$ and $\phi = \log 2$; so that none of them lie in the interval from $\psi = +\pi$ to $\psi = -\pi$. The function $\sigma + \tau i$ is here continuous.

Along the wall

$$\sigma + \tau i = \pm Ai \left\{ \sqrt{2e^{\phi} - e^{2\phi}} - 2 \arcsin \left[\frac{1}{\sqrt{2}} e^{\frac{1}{2}\phi} \right] \right\}.$$

If $\psi < \log 2$ the whole value is purely imaginary, so that $\sigma = 0$, while $\frac{d\tau}{d\phi}$ assumes the value given in (3c). This portion of the lines $\psi = \pm \pi$, therefore, corresponds to the free portion of the stream.

If $\phi > \log 2$, the whole expression as far as $+Ai\pi$ becomes real, which latter belongs to the value of τi through its relation to yi .

The equations (3a) and (3d) correspond therefore to a pouring out of a boundless trough into a canal bordered by two planes, the breadth of the canal being $4A\pi$, and the length of its walls being from $x = -\infty$ to $x = -A(2 - \log 2)$. The free divisional line of the current liquid is at first bent from the edge of the orifice a little towards the side of the positive value of x , where for $\phi = 0$, $x = -A$, and $y = \pm A(\frac{3}{2}\pi + 1)$ it attains its greatest x -value. It then enters the interior of the canal, and at last approaches the two lines $y = \pm A\pi$ asymptotically; so that, finally, the breadth of the stream as it flows out is only equal to half the breadth of the canal.

The velocity along the divisional surface and at the straight end of the stream flowing out is $\frac{1}{A}$. Along the rigid wall, and in the interior of the liquid, the velocity is always less than $\frac{1}{A}$, so that this form of motion may take place with all values of velocity of efflux.

One especially remarks in these examples how it is shown that the form of the liquid current in a tube may be determined for a long way by the shape of the first portion.

Addition, relative to electrical dispersion.—If in equation (2) we look upon the value ψ as the power of the electricity, we get the distribution of the electricity in the neighbourhood of the edge of two plane and almost touching screens under the condition that the distance between the two may be considered vanishingly small compared with the radius of curvature of their curved edges. This is a very simple solution of the problem which Clausius* has discussed. The same distribution of electricity is obtained as was found by Clausius, as far, at least, as it is independent of the curvature of the edge.

I may add that the same method suffices to determine the distribution of the electricity on two parallel infinitely long plane strips whose four corners (in section) form the points of a rectangle. The potential function ψ is given by an equation of the form

$$x + yi = A(\phi + \psi i) + B \frac{1}{H(\phi + \psi i)}, \quad . \quad . \quad . \quad (4)$$

where $H(u)$ represents the function developed by Jacobi in the *Fundamenta Nova*, p. 172, of the numerator of $\sin am u$. The

* Pogg. *Ann.* vol. lxxxvi.

strips correspond, according to the system there adopted, to the values $\phi = \pm 2K$, where $x = \pm 2AK$ gives half the distance of the strips, while the breadth of the strips depends upon the relation between the constants A and B.

From the form of the equations (2) and (4) we may see that ϕ and ψ can only be expressed as functions of x and y by exceedingly complicated serial development.

XLIV. On Unitation ; a novel Arithmetical Operation.

By W. H. WALENN, F.C.S.*

IT has long been known that the remainder to the division of any number by 9 may be determined by dividing the sum of its digits by 9. The following theorem is a generalization and explanation of this property, and establishes a similar operation for all other divisors less than 10.

The theorem is, that if t be the tens' and u the units' digit of a two-figure number, and δ be any integer less than 10, then

$$\overline{10 - \delta t + u}$$

has the same remainder to δ as $10t + u$.

For $\overline{10 - \delta t + u} = 10t + u - \delta t$, and the latter expression is only different from $10t + u$ by an exact number of times δ .

Further, if the digits be s, t, u , the same is true of

$$\overline{10 - \delta^2 s} + \overline{10 - \delta t + u},$$

and so on. For, each time 10 occurs as a factor in any term, it must be treated in the way above indicated, and

$$\overline{10 - \delta^2 s} + \overline{10 - \delta t + u} = \overline{10 - \delta} \overline{10 - \delta s} + \overline{10 - \delta t + u}.$$

These formulæ show the means of determining the remainder to any digit without knowledge of any multiple of that digit, inasmuch as the operation shown by the formula $\overline{10 - \delta t + u}$ only involves multiplication and addition (when $\delta = 9$, addition only); instead of dividing the result of the said operation by δ to obtain the final result, the operation is repeated until only one digit, necessarily one or more units, is obtained. The fitness of regarding this method of obtaining the remainder to δ as an operation totally distinct from that of division, and the fact of single digits only being used, make it convenient to call the said operation "unitation," the remainders being "unitates," and the divisor the "base."

In practice, unitation itself may be simplified by commencing

* Communicated by the Author.